SE 216 – SOFTWARE PROJECT MANAGEMENT
REQUIREMENTS DOCUMENT

PROJECT NAME: REPLATE

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REQ. #	FUNCTIONAL REQUIREMENTS
1	The system should have separate log in forms for municipalities, business owners and clients.
2	Users should be able to register and log in to the application using their email address or phone number, and create a personal profile.
3	The application must display surplus food products on an interactive map, showing the user's location and nearby branches with available products.
4	Users should be able to filter products by category, price range, distance, expiration date, and stock quantity both on the map and list views.
5	For each listed product, the system should display product name, category, original price, discounted price, expiration date, available quantity, branch name, and pickup time window.
6	Users should be able to reserve a product through the application, and the system must generate a unique reservation code and QR code for pickup verification.
7	Once a reservation is made, the reserved product quantity should be deducted from the available stock and the listing should be updated in real time.
8	Users must be able to cancel their reservation up to 30 minutes before the pickup time, and the cancelled quantity should be made available again for other users.
9	The application should send push notifications to users about new surplus food at nearby branches, products approaching their expiration date, price reductions, and donation campaigns.

10	The system should track and display each user's personal impact dashboard showing total food saved (kg), total money saved (TL), and estimated CO2 emission reduction.
11	Businesses should be able to register on the platform with a verification mechanism and declare surplus food by entering product name, category, quantity, expiration date, photo, and estimated pickup time. System will ask automatically if there is an unregistered product with a close expiration date.
12	The system must support a two-stage redistribution model: in the first stage, users can reserve and collect food directly from the restaurant within a limited time window. If the product is not collected, it is automatically transferred to the second stage where it is allocated to municipal distribution centers for further redistribution.
13	Municipal distribution centers shall be able to view incoming product transfers from businesses, track their current stock, and process user pickups by scanning the reservation QR code.
14	The system should allow businesses to view a dashboard with statistics on their surplus food declarations, waste reduction rates, and donation contributions.
15	The system should automatically detect products within 24 hours of their expiration date and apply dynamic pricing suggestions to encourage faster sales.
16	Users should be able to add branches to their favorites list and receive priority notifications when those branches declare new surplus products.
17	The application must allow users to donate meals for people in need through municipal social aid centers, and donate pet food for stray animals to be delivered to animal shelters.
18	Municipalities must have an administrative panel to monitor reported surplus food, track transfers to distribution centers, manage redistribution operations,and analyze regional food waste data.

19	The system may optionally support in-app payment processing for discounted surplus food purchases using secure payment methods.
20	The system must automatically update the status of products (e.g., available, reserved, picked-up, expired, donated, bio-waste) based on user actions and time limits.
REQ. #	**NON-FUNCTIONAL REQUIREMENTS**
1	The system should return product search and filter results in less than 3 seconds under normal network conditions.
2	The system should maintain at least 99% uptime availability and be accessible 24/7.
3	The system should be able to restart itself within 5 minutes in case of a crash or unexpected failure.
4	All user personal data, payment information, and reservation data must be stored securely using encryption (AES-256) and transmitted over secure protocols (SSL/TLS).
5	Only authorized administrators should have access to system management panels and sensitive business data.
6	The application should provide a user-friendly and intuitive interface so that users of all ages can easily search for products and complete reservations.
7	The system should be scalable to support a growing number of users, businesses, and branches without performance degradation.
8	The system should perform automated daily data backups to prevent data loss in case of system failure.
9	The application should be compatible with both iOS and Android platforms, and should function properly on devices with different screen sizes and hardware specifications.

10	The system should be developed using a modular architecture so that new features such as AI-based demand forecasting and pet food donation can be integrated without major refactoring.
11	The map and location-based features should accurately detect the user's GPS location with a maximum error margin of 25 meters.
12	Regular maintenance and updates must be provided to keep the application running smoothly, and critical security patches should be applied within 48 hours of discovery.